

Multiple Choice (10 marks)

1. The walls of bronchioles contain
 - (a) cartilage and respiratory epithelium.
 - (b) Only respiratory epithelium
 - (c) Only cartilage
 - (d) Respiratory epithelium and cartilage
 - (e) smooth muscle and respiratory epithelium.
2. The creatine kinase reaction is:
 - (a) constantly activated, regardless of ATP levels.
 - (b) more efficient than glycolysis.
 - (c) slow compared with glycolysis.
 - (d) not involved in energy production.
 - (e) irreversible.
3. Which of the following disorders does not show X-linked inheritance?
 - (a) Haemophilia A
 - (b) Fragile X syndrome
 - (c) Becker muscular dystrophy
 - (d) Color blindness
 - (e) Lesch-Nyhan syndrome
4. How are new polyomaviruses detailed
 - (a) Shot gun sequencing
 - (b) Cultivation in human neural cells
 - (c) Deep pyro sequencing (NGS)
 - (d) Monoclonal antibody techniques
5. Which products of ADP degradation increase in concentration in the blood during multiple sprint sports?
 - (a) Ammonia, urea and creatine.
 - (b) Ammonia, hypoxanthine and creatinine.
 - (c) Ammonia, creatine and uric acid.
 - (d) Urea, creatine and uric acid.
 - (e) Ammonia, urea and uric acid.

True / False (10 marks)

Answer True or False

6. True or False: Older adults typically excel in tests of episodic memory, which involves personal experiences and specific events.
7. True or False: Biotin plays a crucial role in the carboxylation of pyruvate to form oxaloacetate, an essential step in gluconeogenesis.
8. True or False: Vaccines have completely stopped the spread of all infectious diseases worldwide, eliminating the need for further public health measures.
9. True or False: Measuring blood pressure in an arm above the heart level will accurately reflect the true diastolic pressure.
10. True or False: First cousins share, on average, one-eighth of their genes, reflecting a closer genetic linkage than actual.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of a relative risk ratio of 1.2 for hip fractures among elderly women who exercise regularly, considering its statistical significance and potential impact on public health recommendations.
12. Discuss the role of dietary saturated fat in the etiology of lung cancer, analyzing its potential mechanisms and implications for public health strategies.

End of Paper